

Dupire PINN / PDF validation — canonical journal

Dupire local volatility via two PINNs: a journal of work

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JCF manuscript — LocalVolatility PINN pipeline | canonical reference (supersedes
main / density_story / dec10_collate)

2025-11-02 → 2026-06-03

Outline

Results first

Timeline & conventions

Model & validation pipeline

Synthesis & open decisions

How to read this journal

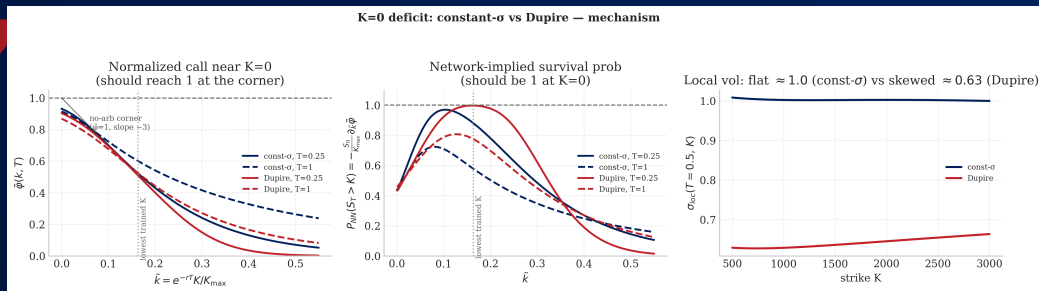
- §1 is the answer. Headline results and figures come *first*, before any derivation.
- §§4–8 are the chronology, oldest → newest; each opens with a dated divider. §2 carries the master timeline.
- Appendices hold the derivations, full per-model tables, DAX details and a provenance changelog.
- All numbers are ground-truth from committed diagnostics (`plots/nn_k0_check/*.json`, `plots/reproduce_user_regen/*/diagnostics.json`). Symbols are defined once on the Conventions slide (§2).
- This deck merges and supersedes `main.tex`, `density_story.tex`, `dec10_collate.tex`; those are retained unedited for history.

The verdict, in one paragraph

The two networks calibrate well inside the traded strike range. A single structural defect — the price net NN_φ cannot represent $\tilde{\varphi} = 1$ at $K = 0$ — produces a maturity-growing price deficit. The volatility net NN_η is sound.

- Direct $K = 0$ call deficit $1 - \tilde{\varphi}(0, T)$: **const- σ 6.85 $\rightarrow$ 8.53%, Dupire 9.65 $\rightarrow$ 13.15%** ($T = 0.25 \rightarrow 1.0$). Dupire $\sim 1.5\times$ worse.
- Root cause is localized to NN_φ at $K = 0$ by an integration-by-parts (IBP) three-way mean test *and* a data-vs-repriced MC check (NN_η tight to ≤ 0.4 pp).
- A $K = 0$ boundary penalty (retrain) brings all three IBP legs **within $\sim 1\%$** .

Headline figure: three $K = 0$ mechanisms at a glance



L: normalized call $\tilde{\varphi}$ rounds off below the no-arbitrage corner ($\tilde{\varphi}=1$, slope -3), Dupire lower. **M:** the network-implied survival probability $P_{NN}(S_T > K)$ is ≈ 1 at the data edge but **collapses to ≈ 0.45 at $K=0$** . **R:** local vol — flat ≈ 1.0 (const- σ) vs mildly skewed ≈ 0.63 (Dupire).

Headline table: NN call at $K = 0$ (the two paper models)

model	T	$\mathcal{N}_c(0, T)$	$\tilde{\varphi}(0, T)$	$C_{\text{NN}}(0, T)$	deficit $1 - \tilde{\varphi}$
const- σ ($\sigma=1$)	0.25	2.681	0.9315	931.5	6.85%
	0.50	2.632	0.9280	928.0	7.20%
	0.75	2.555	0.9223	922.3	7.77%
	1.00	2.462	0.9147	914.7	8.53%
Dupire	0.25	2.338	0.9035	903.5	9.65%
	0.50	2.255	0.8952	895.2	10.48%
	0.75	2.143	0.8827	882.7	11.73%
	1.00	2.029	0.8685	868.5	13.15%

By no-arbitrage $C(0, T) = S_0 = 1000$ ($\tilde{\varphi} = 1$). The trained net plateaus at a *finite* $\mathcal{N}_c$, so $\tilde{\varphi}(0, T) < 1$. Source: `nn_k0_check/{const_vol,dupire}/nn_at_k0.json`.

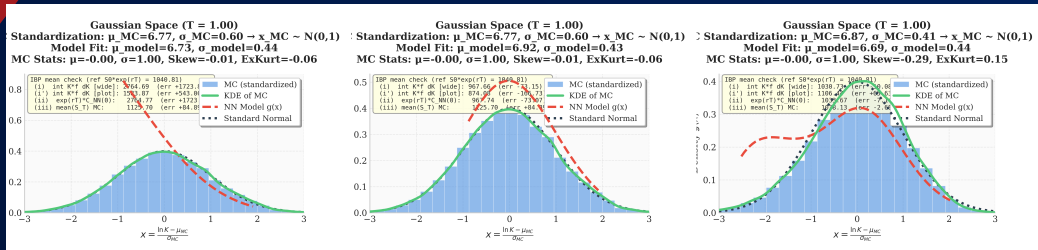
Headline: IBP three-way, before vs after the $K = 0$ fix

Legs all equal the forward $M_{\text{ref}} = S_0 e^{rT}$ under no-arbitrage ($S_0=1000$, $r=0.04$):

model	T	M_{ref}	(i)= $e^{rT} \int K \partial_K^2 C$	(ii)= $e^{rT} C_{\text{NN}}(0)$	(iii)= $\mathbb{E}[S_T]$
example_pretrained	0.5	1020.20	948.86	948.93	$\approx 7\%$ low
	1.0	1040.81	967.65	967.74	
	1.5	1061.84	982.49	982.95	
k0_bc_retrain ($\lambda_{k_0}, K_{\text{min}}=100$)	0.5	1020.20	1012.20	1014.34	1018.92
	1.0	1040.81	1030.73	1033.67	1038.13

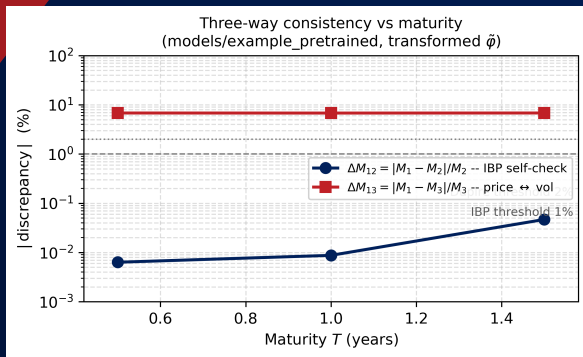
(i) $\equiv$ (ii) to $\sim 6 \times 10^{-5}$ (IBP/autodiff exact). Pretrained $\sim 7\%$ below forward; the $K = 0$ penalty retrain pulls all three legs to **within $\sim 1\%$** .

Headline: before $\rightarrow$ after at $T = 1.0$



Gaussian-space density: legacy $\tilde{\varphi}$ bug $\rightarrow$ corrected pretrained $\rightarrow$ retrained with $K = 0$ boundary condition. The boundary is progressively pinned.

Headline: ΔM_{13} vs maturity — corrected per model

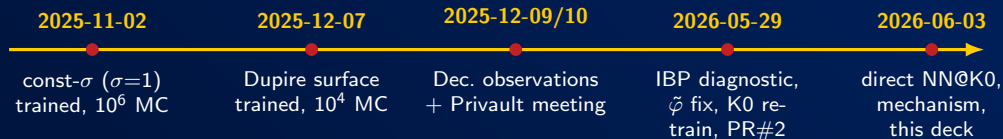


$\Delta M_{13} = |M_1 - M_3|/M_3$ (data MC):

T	const- σ	Dupire
0.25	6.77%	9.63%
0.50	7.11%	10.44%
0.75	8.00%	11.72%
1.00	9.99%	13.13%

Correction: dec10_collate captioned the Dupire case with const- σ numbers ($\sim 6.8\%$); the true Dupire gap is **9.6–13.1%**.

Timeline of the work



Each chronology section (§§4–8) opens with a dated divider keyed to one node above.

Conventions & symbols (the single source of truth)

Scaling (as the network is trained):

$$t = \frac{T}{T_{\max}}, \quad k = \frac{e^{-rT} K}{K_{\max}}, \quad \tilde{\varphi} = \frac{C}{S_0}, \quad \eta = \frac{1}{2} T_{\max} \sigma^2, \quad \boxed{\tilde{\varphi} = 1 - e^{-\mathcal{N}_c}} \quad (\mathcal{N}_c = \text{NN}_{\varphi} > 0).$$

Fixed constants: $S_0 = 1000$, $r = 0.04$, $K_{\max} = 3000$, $T_{\max} = 1.5$, training $K \in [500, 3000]$.

The two synthetic models (do not mix):

- **const- σ :** $\sigma(x, t) \equiv 1$ (constant), $r = 0.04$, 10^6 MC paths. (*Config default sigma_base=0.3 is unused — the custom function returns 1.*)
- **Dupire:** $\sigma(x, t) = 0.3 + y e^{-y}$, $y = (t+0.1)\sqrt{x+0.1}$, $r = 0.04$, 10^4 MC paths.
- A separate *December control* used $\sigma=1$ with $r=0.02$ — flagged where it appears.

Mean estimators & deltas: $M_1 = e^{rT} \int K \partial_K^2 C$ (density), $M_2 = e^{rT} C(0, T)$ (call@0), $M_3 = \mathbb{E}[S_T]$ (**unconditional** MC mean); $\Delta M_{12} = \frac{|M_1 - M_2|}{M_2}$, $\Delta M_{13} = \frac{|M_1 - M_3|}{M_3}$. (Trimmed in-window M_3 biases up $\sim 2\%$ — caveat in Appendix B.)

The two networks

- NN_φ (price): outputs $\mathcal{N}_c$, giving the call surface $C(T, K) = S_0 \tilde{\varphi} = S_0(1 - e^{-\mathcal{N}_c})$.
- NN_η (volatility): outputs η , hence $\sigma = \sqrt{2\eta/T_{\max}}$, feeding the Dupire PDE $\partial_t \varphi = \eta k^2 \partial_k^2 \varphi$.
- Each: GaussianNoise $\rightarrow$ Dense(64, tanh) $\rightarrow$ 3 residual blocks $\rightarrow$ Dense(1, softplus); scaled inputs (t, k) .

Forward pointer: NN_η is verified sound (§8 interchangeability + M_3 on the forward); NN_φ at $K = 0$ is the locus of the defect.

Validation toolkit

- **PDF triptych** per maturity: strike-space $f(K)$, log-strike $f(\ln K)$, Gaussian (standardized MC KDE vs NN, against $\mathcal{N}(0, 1)$).
- **Breeden–Litzenberger**: $q(K, T) = e^{rT} \partial_K^2 C = f_Q(K)$ (derivation in Appendix A).
- **IBP three-way mean test**: three independent estimates M_1, M_2, M_3 that must all equal the forward $S_0 e^{rT}$ — the sharpest localizer of where a deviation lives.



2025-11-02

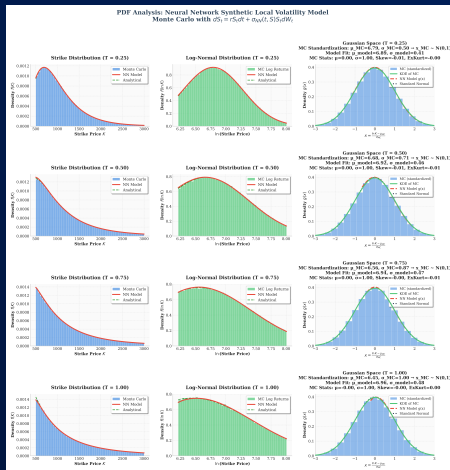
Constant- σ synthetic model

$\sigma=1$, $r=0.04$, 10^6 MC paths $\rightarrow$ model ..._constant_vol

Const- σ setup

- Model $dS_t = rS_t dt + \sigma S_t dW_t$, $\sigma = 1$ constant, $r = 0.04$, $S_0 = 1000$.
- Grid $K \in [500, 3000]$, $T \in \{0.25, 0.5, 0.75, 1.0\}$; benchmark = closed-form log-normal.
- Trained from 10^6 MC paths ($\ln 10^{-3}$). *Distinct from the December $\sigma=1, r=0.02$ control (§6).*

Const- σ density: the clean baseline



NN, MC and the log-normal analytical density are essentially indistinguishable; σ_{MC} tracks $\sigma_{\text{truth}}\sqrt{T}$.
The reference behaviour — no deviation.

Const- σ : excellent in-sample, so $K = 0$ is extrapolation

- In-sample fit at the lowest trained strike $K = 500$ ($k \approx 0.16$): **0.07% mean** (max 0.21%) error against the data — the model is excellent where it has data.
- Therefore the $K = 0$ deficit is *not* underfitting; it is extrapolation below $K_{\min}$ (quantified in §8). $C_{\text{NN}}(0, T)$ lands $6.85 \rightarrow 8.53\%$ below S_0 .



2025-12-07

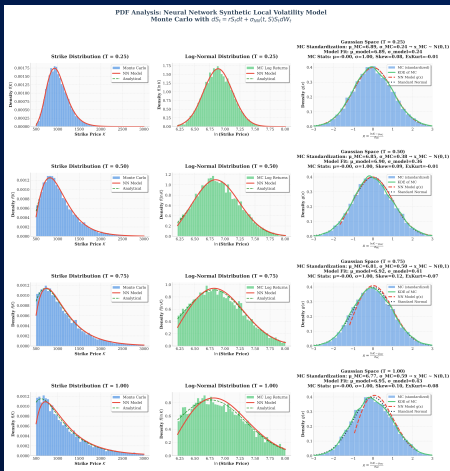
Dupire synthetic model

$\sigma(x,t) = 0.3 + y e^{-y}$, $r=0.04$, 10^4 MC paths (100× fewer) →
..._dupire_exact

Dupire setup

- Local vol $\sigma(x, t) = 0.3 + y e^{-y}$, $y = (t+0.1)\sqrt{x+0.1}$ (manuscript §4.1); $r = 0.04$, $S_0 = 1000$.
- Same grid $K \in [500, 3000]$, $T \in \{0.25, \dots, 1.0\}$. The NN must recover the non-flat σ^2 from prices alone.
- Trained from 10^4 MC paths ($\text{lr } 10^{-4}$) — 100× fewer paths than const- σ .

Dupire density: skew recovered, small deficit emerging



NN tracks the non-log-normal MC closely; a small deficit is already visible in the Gaussian panel.
Recovered local vol ≈ 0.63 — *lower* than const- σ 's 1.0.

Dupire: the seed of the larger gap

- In-sample fit at $K = 500$: 0.21% mean (max 0.55%) — still good, $3\times$ const- σ .
- Lower vol (≈ 0.63 vs 1.0) $\Rightarrow$ narrower, steeper terminal law $\Rightarrow$ a sharper $K = 0$ corner.
- Foreshadow: this is exactly why the $K = 0$ deficit is larger here ($9.65 \rightarrow 13.15\%$) — mechanism quantified in §8.



2025-12-09/10

December observations + Privault meeting

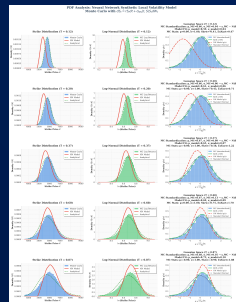
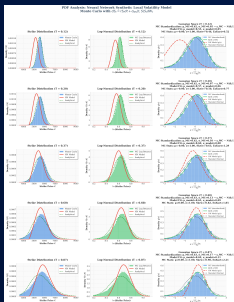
KDE-vs-NN deviation near $K=0$ first seen, across synthetic and DAX market data

What December looked at (and the $\sigma=1$ control)

- PDF/density analysis of both synthetic models plus a $\sigma=1$, $r=0.02$ control (kept distinct from the 2025-11 paper model, $r=0.04$).
- Three-panel densities flagged a systematic NN-vs-MC deviation near $K = 0$ in Gaussian space.
- Presented to N. Privault on Dec 10; his integration-by-parts mean test became the diagnostic (§7).

Case 3: DAX market data (day mapping confirmed)

- DAX index options, 3 consecutive days, Aug 2001; $K \in [4000, 9000]$, $T \in \{0.12, 0.20, 0.37, 0.60, 0.87\}$.
- **Confirmed spot S_0 :** 5752.51 (7 Aug), 5614.51 (8 Aug), 5512.28 (9 Aug); source `examples/dax_analysis_common.py` (`date_label` fields).
- The earlier “provisional day assignment” caveat is withdrawn — the mapping is verified.



What December established

- The KDE-vs-NN deviation near $K = 0$ is **real and structural** — it appears in synthetic const- σ , synthetic Dupire *and* DAX market data (3 days, Fig. above).
- Open question entering 2026: *where* does it come from — the price net, the vol net, or the diagnostic itself? $\Rightarrow$ the IBP three-way test.

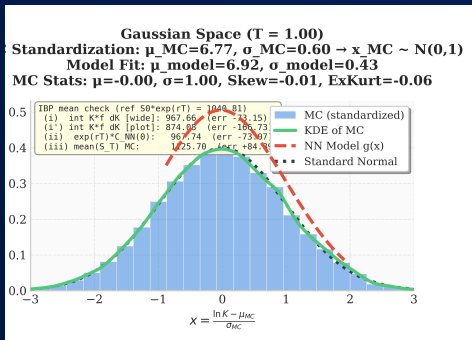


2026-05-25 → 05-29

IBP diagnostic & the $\tilde{\varphi}$ bug

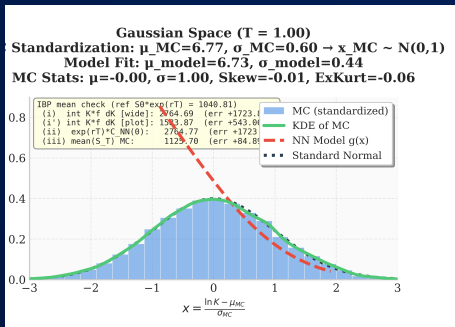
three-way mean test; the $(1 - k)$ factor; $K=0$ penalty retrain; PR#2
merged 05-29

What Nicolas saw → the IBP question



The mean identity $e^{rT} \int K \partial_K^2 C = e^{rT} C(0, T) = \mathbb{E}[S_T]$ must hold. If the means agree but the densities differ, *where* is the deviation? The three-way test answers it.

The $\tilde{\varphi}$ bug: raw NN_{φ} used as $\tilde{\varphi}$



Using the raw network output as $\tilde{\varphi}$ (instead of $1 - e^{-\mathcal{N}_c}$) inflated legs (i)/(ii) by $\sim 2.7\times$. This was a *diagnostic-code* bug, not a model bug; fixed 05-25 $\rightarrow$ 29. Correct map: $\tilde{\varphi} = 1 - e^{-\mathcal{N}_c}$.

Three-way result: (i) $\equiv$ (ii), M_3 on the forward

example_pretrained, synthetic surface:

T	M_{ref}	M_1	%err	M_2	%err	ΔM_{12}	ΔM_{13}
0.5	1020.20	948.86	-6.99	948.93	-6.99	0.006%	6.81%
1.0	1040.81	967.65	-7.03	967.74	-7.02	0.009%	6.79%
1.5	1061.84	982.49	-7.47	982.95	-7.43	0.047%	6.82%

- $M_1 \equiv M_2$ to $< 0.05\%$: the IBP / autodiff / quadrature chain is exact.
- M_3 within 0.7% of the forward: NN_η is well-calibrated.
- $M_1, M_2 \approx 7\%$ below forward: NN_φ undershoots near $K = 0$.

The $(1 - k)$ discrepancy, stated in full

A genuine **manuscript-vs-code** difference in the price ansatz:

$$\text{Manuscript (Eq. 3.9): } \tilde{\varphi} = 1 - e^{-(1-k)\mathcal{N}_c}, \quad \text{Code (as trained): } \tilde{\varphi} = 1 - e^{-\mathcal{N}_c}.$$

- The $(1 - k)$ factor pins the *far* boundary: at $k = 1$ ($K = \infty$) it forces $\tilde{\varphi} = 0$ exactly, by construction.
- At the *near* boundary $k = 0$ ($K = 0$), $(1 - k) = 1$ — the factor is inert there.
- So it cannot be the cause of, or the fix for, the $K = 0$ deficit. Next slide: the counterfactual.

$(1 - k)$ **counterfactual**: it does not fix $K = 0$

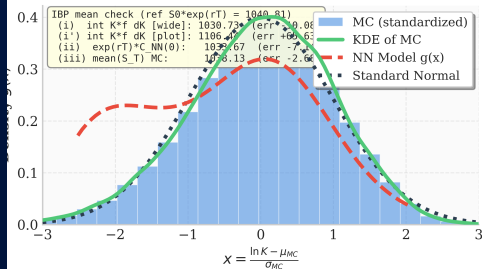
Applying Eq. 3.9's $(1 - k)$ factor to the trained $\mathcal{N}_c$ at analysis time:

T	M_2	M_1^{cap}	$M_1(\text{pred})$	$\Delta M_{12}^{\text{cap}}$
0.5	948.93	948.25	948.22	0.072%
1.0	967.74	954.37	954.42	1.381%
1.5	982.95	949.51	949.50	3.402%

- $k = 0$: M_2 **unchanged** (factor = 1) — the deficit is identical with or without $(1 - k)$.
- $k = 1$: $\tilde{\varphi}$ forced to exactly 0.
- But it **introduces** a new $M_1 \neq M_2$ slope gap $= e^{rT} S_0 \mathcal{N}_c(1)$, growing $0.07 \rightarrow 1.4 \rightarrow 3.4\%$ with T .
- **Verdict**: $(1 - k)$ is a far-boundary cosmetic; the $K = 0$ deficit is untouched.

The fix: $K = 0$ penalty + wider strikes

Gaussian Space ($T = 1.00$)
 Standardization: $\mu_{MC}=6.87, \sigma_{MC}=0.41 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=6.69, \sigma_{model}=0.44$
 MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.29, \text{ExKurt}=0.15$



Add $\lambda_{k_0} |C(0, T) - S_0|^2$ to the loss; train strikes down to $K_{\min} = 100$ ($\lambda_{k_0}=10$, 3000 epochs) $\rightarrow$ k0_bc_retrain.

At $T=1.0$: (i)= 1030.7, (ii)= 1033.7, (iii)= 1038.1, ref= 1040.8 — within $\sim 1\%$.

PR#2 merged to main on 2026-05-29.

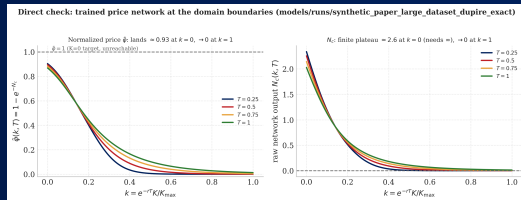
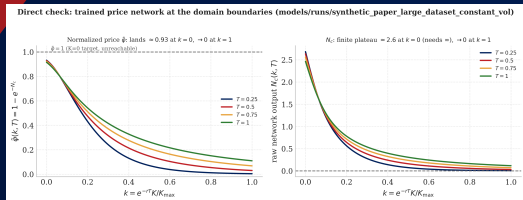


2026-06-02/03

**This session: direct $K=0$ evaluation +
mechanism**

NN read off at the boundary; survival-probability collapse; data-vs-repriced interchangeability

Direct NN at $K = 0$ — both paper models



Left const- σ , right Dupire. $\tilde{\varphi}(k) = 1 - e^{-N_c}$ lands at $\approx 0.91/0.87$ at $k=0$ (not 1); $\rightarrow 0$ cleanly at $k=1$. $N_c(0)$: const- σ 2.46–2.68, Dupire *lower* 2.03–2.34 — and deficit $= e^{-N_c}$, so Dupire's lower plateau $\Rightarrow \sim 1.5\times$ larger deficit.

Mechanism 1 — $K = 0$ is extrapolation, not misfit

- Training strikes $K \in [500, 3000] \Rightarrow k \geq 0.16$; $K = 0$ is $\sim 16\%$ of the domain *below* any datum — the **same extrapolation distance for both models**.
- In-sample fit at $K = 500$: const- σ 0.07%, Dupire 0.21% (mean) — both excellent.
- So the deficit appears only where there is no data, and (for these two models) no $K = 0$ boundary constraint.

Mechanism 2 — the implied survival probability collapses

No-arbitrage at $K = 0$: $\tilde{\varphi}(0) = 1$ with slope $\partial_k \tilde{\varphi}(0) = -K_{\max}/S_0 = -3$, i.e. implied survival $P_{NN}(S_T > K) = -(S_0/K_{\max}) \partial_k \tilde{\varphi} = 1$.

- **Measured** $\partial_k \tilde{\varphi}(0) \approx -1.33$ (less than half the required -3) $\Rightarrow P_{NN}(S_T > 0) \approx 0.45$ for *both* models.
- Equivalently, the extrapolated surface implicitly puts $\sim 55\%$ spurious probability on $S_T \leq 0$ — and the softplus head cannot deliver $\tilde{\varphi} = 1$.
- $P_{NN} \approx 1$ right at the data edge ($k \approx 0.16$) and collapses only in the extrapolation region (Fig., §1 middle panel) — the smoking gun.

Mechanism 3 — why Dupire is worse, and the T -trend

- **Why worse:** Dupire's local vol (≈ 0.63) is lower than const- σ 's (1.0) $\Rightarrow$ narrower terminal law $\Rightarrow$ sharper $K = 0$ corner $\Rightarrow$ lower $\mathcal{N}_c$ plateau; $e^{-\mathcal{N}_c}$ then gives the $\sim 1.5\times$ gap. Looser calibration (10^4 vs 10^6 paths, $10\times$ lower lr) pushes the same way.
- **T -trend (a nuance):** the scaled Dupire PDE $\partial_t \varphi = \eta k^2 \partial_k^2 \varphi$ has the k^2 factor vanish at $k = 0$, so in the idealized limit $\tilde{\varphi}(0, t)$ is *frozen* in T . In practice the trained $\mathcal{N}_c(0, T)$ drifts down with T , so the realized deficit grows mildly (e.g. const- σ $6.85 \rightarrow 8.53\%$).

Interchangeability: the gap is NN_φ , not the σ surface

- Substitute the MC mean (M_3) between **data MC** (10^6 paths, ground-truth σ) and **repriced MC** (10^5 paths, learned σ_{NN}).
- M_3 changes $\leq 0.5\%$ and $\Delta M_{13} \leq 0.4 \text{ pp}$ at every (model, T).
- $\Rightarrow \text{NN}_\eta$ reproduces each σ surface tightly; the residual lives entirely in NN_φ at $K = 0$, not in the vol leg or the $\partial_K^2 C$ chain.
- Practical bonus: the cheap 10^5 -path repriced MC (~ 2 min) is interchangeable with the 10^6 -path data MC for these diagnostics.

ΔM_{13} reconciliation — the IBP gap is the $K = 0$ deficit

Paper-model IBP three-way (data MC, unconditional M_3):

model	T	M_1	M_2	M_3	M_{ref}	ΔM_{13}
const- σ	0.25	940.77	940.86	1009.06	1010.05	6.77%
	1.00	935.49	952.05	1039.26	1040.81	9.99%
Dupire	0.25	912.45	912.54	1009.63	1010.05	9.63%
	1.00	903.78	903.92	1040.39	1040.81	13.13%

- ΔM_{13} matches the direct $K = 0$ deficits (§1 table) to < 0.5 pp — the IBP gap and the boundary shortfall are the same effect.
- const- σ shows a small $M_1 \neq M_2$ at $T=1.0$ (935.5 vs 952.1, a real ΔM_{12}); Dupire's two legs are tight.

What is settled

1. The defect is structural: NN_φ cannot reach $\tilde{\varphi} = 1$ at $K = 0$ (softplus range open at 1).
2. NN_η is sound: interchangeability (≤ 0.4 pp) and M_3 on the forward.
3. The $(1 - k)$ factor does not fix $K = 0$ (inert there); it only pins $K = \infty$.
4. A $K = 0$ penalty retrain brings all three IBP legs within $\sim 1\%$.
5. DAX day-to-file mapping is confirmed (no longer provisional).

Open decisions — to resolve here, together

- **Manuscript Eq. 3.9:** keep $(1 - k)$ for the far boundary but *add* a $K = 0$ mechanism? Or adopt the parity ansatz $\pi^c = S_0 - Ke^{rT} [1 - (1 - k)e^{-\mathcal{N}_c}]$ that pins *both* ends by construction?
- **Paper model:** ship `k0_bc_retrain`, or retrain Dupire with more paths / higher lr ?
- **M_3 convention:** freeze to the unconditional MC mean for all manuscript tables (recommended).
- **Co-author sign-off:** confirm the $(1 - k)$ discrepancy is documented here as agreed.

Questions & discussion

Canonical deck — supersedes `main.tex`, `density_story.tex`, `dec10_collate.tex`.

Diagnostics: `plots/nn_k0_check/` | IBP: `plots/ibp_comparison/` | math:
`docs/MATHEMATICAL_TREATMENT.md`

$$C(K, T) = e^{-rT} \int_K^\infty (S - K) f(S) dS, \quad \frac{\partial C}{\partial K} = -e^{-rT} \mathbb{Q}(S_T > K), \quad \boxed{q = e^{rT} \partial_K^2 C = f_{\mathbb{Q}}}.$$

$$\mathbb{E}[S_T] = e^{rT} \int_0^\infty K \partial_K^2 C dK = e^{rT} [K \partial_K C - C]_0^\infty = e^{rT} C(0, T) = \boxed{S_0 e^{rT}}.$$

Scaled (k linear in K , constant Jacobian, no curvature term): $q = \frac{S_0 e^{-rT}}{K_{\max}^2} \partial_k^2 \tilde{\varphi}$. This is exactly what `_raw_model_density` computes (double autodiff $\times$ Jacobian² $\times S_0 \times e^{rT}$); $M_1 \equiv M_2$ to machine precision.

Appendix B — M_3 : unconditional vs in-window (trimming caveat)

- The IBP three-way uses the **unconditional** MC mean for M_3 .
- An earlier diagnostic used the $[K_{\min}, K_{\max}]$ -trimmed in-window mean, which under local vol has a heavy right tail and biased M_3 up $\sim 2\%$ (e.g. 1042.19 vs 1018.25 at $T=0.5$ for `example_pretrained`); patched at `dupire_pipeline.py` (full-sample mean).
- Manuscript tables should standardize on the unconditional mean.

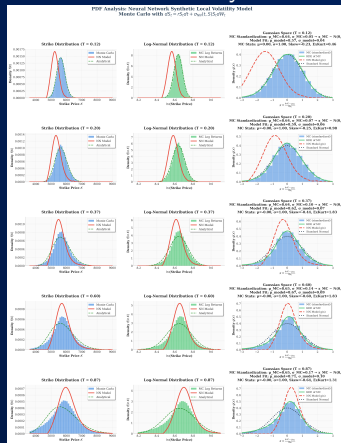
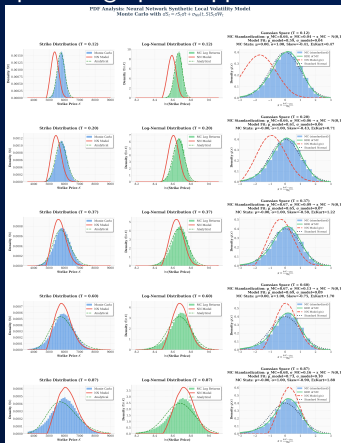
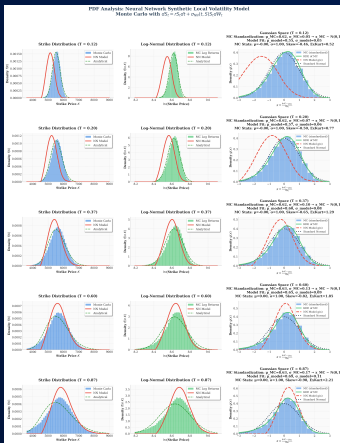
Appendix C — full per-model boundary tables

model	T	$k_{\min}$	$\tilde{\varphi}(k_{\min})$	$\tilde{\varphi}(0)$	$\partial_k \tilde{\varphi}(0)$	$P_{NN}(S_T > 0)$	deficit
const- σ	0.25	0.165	0.518	0.9315	-1.363	0.454	6.85%
	0.50	0.163	0.550	0.9280	-1.332	0.444	7.20%
	0.75	0.162	0.581	0.9223	-1.326	0.442	7.77%
	1.00	0.160	0.607	0.9147	-1.332	0.444	8.53%
Dupire	0.25	0.165	0.505	0.9035	-1.336	0.445	9.65%
	0.50	0.163	0.512	0.8952	-1.365	0.455	10.48%
	0.75	0.162	0.522	0.8827	-1.390	0.463	11.73%
	1.00	0.160	0.532	0.8685	-1.406	0.469	13.15%

No-arbitrage requires $\partial_k \tilde{\varphi}(0) = -3$, $P_{NN} = 1$. At $k=1$, $\mathcal{N}_c \approx 0.004\text{--}0.13$, $\tilde{\varphi} \rightarrow 0$ — the far boundary is reached. Source: `nn_k0_check/*/nn_at_k0.json`.

Appendix D — DAX details (confirmed)

- Spot S_0 : 5752.51 (7 Aug 2001), 5614.51 (8 Aug), 5512.28 (9 Aug); source of truth examples/dax_analysis_common.py.
- $K \in [4000, 9000]$, $T \in \{0.12, 0.20, 0.37, 0.60, 0.87\}$; per-day $\mu_{MC} \approx 8.64\text{--}8.66$ (ln K).
- The same $K = 0$ Gaussian-space fingerprint appears on all three consecutive days.



Appendix E — code, reproduction & provenance

- **Key code:** `dupire_pipeline.py` (`phi_tilde_from_nn`, `_raw_model_density`, `compute_mean_diagnostics`; λ_{k_0} in `loss_phi_cal`); `config.py`; `examples/`: `check_nn_at_k0.py`, `check_k0_discrepancy.py`, `run_ibp_comparison_plots.py`, `run_k0_retrain.py`.
- **Reproduce K=0:** `python examples/check_nn_at_k0.py --model-dir models/runs/..._constant_vol --maturities 0.25 0.5 0.75 1.0`.
- **Compute routing:** analysis local ($\sim$ min); training off-Mac (GPU/PBS).
- **Provenance:** this deck merges `main.tex` (§7 IBP arc), `density_story.tex` (App. A derivation, $(1 - k)$ counterfactual), `dec10_collate.tex` (§§4–6 case studies); new §§1,8 from the 2026-06 session. PR#2 merged 05-29; diagnostics committed 06-02/03.